LEAK OR KEEP?
LEARNING TO SHARE INTERVIEW QUESTIONS AMONG COMPETING CANDIDATES

PROJECT IDEA REVIEW — MULTI-AGENT SYSTEMS, IIT GANDHINAGAR

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ABSTRACT

After an interview, a candidate can tell friends who interview later which questions were asked. This helps them, but they may be competing for the same offers. We model this as a game on an $n \times n$ *sharing matrix*: candidates interview in a fixed order, know only a coarse ability type (never their score), and act in a random environment, since a shared question may not reappear. At Level 1 (known environment) we estimate the payoff table by simulation and solve it for pure, mixed and coarse correlated equilibria. At Level 2 (unknown environment) each candidate is a bandit learner who must learn the random payoff of every sharing decision. We vary the hiring rule, friendship network and utility functions. In a six-candidate pilot (one setting), the equilibrium found under fixed seats has weak candidates sharing and average and strong candidates keeping quiet, whereas under a cut-off everyone shares. After 60,000 rounds, the most-played actions of the learners match this equilibrium in 39–97% of decision cells, depending on the learner and the rule.

Keywords Bayesian games · multi-agent bandits · empirical game theory · coarse correlated equilibrium

1 Introduction

Every placement season, a candidate leaves the interview room and within minutes their friends know what was asked. It feels like helping friends, but when offers are limited, a friend helped may take the helper's seat. Leaks happen at scale [1, 2]; industry treats them as something that *happens to* questions (bigger banks, rotation [2]), not a *choice* of competing candidates.

Example. Two offers; Aarav, Meera, Kabir and Diya interview in that order, and Kabir and Diya are Aarav's friends. Each answers 3 of 10 bank questions, and knowing a question adds 4 of its 10 marks. With scores 21, 22, 20 and 19, Meera and Aarav are hired. If Aarav shares, each friend sees at least one of his questions with probability $1 - \binom{7}{3}/\binom{10}{3} \approx 0.71$, and with probability 0.92 one of them overtakes him. If the company instead hires everyone scoring at least 20, sharing costs Aarav nothing — and Diya, meant to be rejected, gets in with probability 0.71.

Question. *Who shares with whom, how does this depend on the hiring rule and on what candidates care about, and can candidates who do not know the payoffs learn to share as the equilibrium predicts?*

Related work. Rank-order versus absolute rewards [3] and helping or sabotaging rivals [4] are well studied, but there help is effort, not information. Li and Conitzer [5] model leaked test questions as a Stackelberg game with the leak given; we make it a choice among competitors. Unknown payoffs make candidates bandit learners [6–8], whose no-regret play approaches coarse correlated equilibria [9]; solving a simulated game is empirical game-theoretic analysis [10].

2 Preliminaries

Notation. $\mathcal{N} = \{1, \dots, n\}$ are the candidates, numbered in their (fixed) interview order. A symmetric matrix $W \in \{0,1\}^{n \times n}$ records friendships: $F(i)$ is the set of i's friends and $L(i) = \{j \in F(i) : j > i\}$ those who interview *after* i. Candidate i has a skill s_i and a coarse *ability type* $\theta_i \in \Theta = \{\text{weak}, \text{average}, \text{strong}\}$, the tercile of s_i. The company has K seats, a bank of Q questions, and asks m per interview. Y_j is j's score, H the hired set, and $x_j = \mathbb{1}[j \in H]$. The *sharing matrix* $S \in \{0,1\}^{n \times n}$ has $S_{ij} = 1$ if i tells j; row $S_{i\cdot}$ is i's action.

Definition 1 (Bayesian game). A Bayesian game is a tuple $\langle \mathcal{N}, (A_i), (\Theta_i), p, (u_i) \rangle$: players, action sets, type sets, a common prior p over types, and utilities u_i depending on all actions and types [11]. Each player thus holds private information (her ability type), while the type distribution is common knowledge.

Definition 2 (Strategy). A strategy is a map $\sigma_i : \Theta_i \to \Delta(A_i)$ from a player's type to a probability distribution over her actions. It is *pure* if that distribution puts all weight on one action, and *mixed* otherwise. A strategy is thus a rule of the form "given my type, tell these friends", possibly randomised.

Definition 3 (Bayes–Nash equilibrium, BNE). A profile $\sigma = (\sigma_1, \dots, \sigma_n)$ is a BNE if, for every player i, type θ_i and action a_i', $\mathbb{E}[u_i(\sigma_i(\theta_i), \sigma_{-i}) \mid \theta_i] \geq \mathbb{E}[u_i(a_i', \sigma_{-i}) \mid \theta_i]$, where the expectation is over the others' types and the environment. That is, given the others' strategies, no type of any player can gain by changing her own action. The *exploitability* of σ is the largest expected gain any type can obtain by deviating; it is 0 exactly at a BNE, and σ is an ε-BNE if its exploitability is at most ε.

Definition 4 (Coarse correlated equilibrium, CCE). A distribution z over strategy profiles is a CCE if, for every player i and every fixed alternative strategy σ_i', $\mathbb{E}_{\sigma \sim z}[U_i(\sigma)] \geq \mathbb{E}_{\sigma \sim z}[U_i(\sigma_i', \sigma_{-i})]$, where U_i is expected utility over types and the environment. Equivalently, if a coordinator draws a profile from z and each player must decide *before* the draw whether to follow it, no player gains by opting out. Every BNE, pure or mixed, is a CCE, and the CCEs form a polytope, so they can be found by a linear program. A mixed equilibrium can also be read as pure play under small private payoff shocks [12] (*purification*): "random" sharing may be a deterministic rule on private information.

Definition 5 (Contextual bandit and regret). In round t a learner sees a context x_t (here her type θ_i^t), picks an arm a_t (a row of S), and observes *only* the reward r_t of that arm. Its regret after T rounds is $\mathrm{Reg}(T) = \max_\pi \sum_{t \leq T} r_t(\pi(x_t)) - \sum_{t \leq T} r_t(a_t)$, the reward lost against the best fixed rule π from contexts to arms. The learner never observes what the other arms would have paid, so it must trade off trying arms (exploration) against playing the best one so far (exploitation). We use ε-greedy, UCB1 [6], Thompson sampling [8] and EXP3 [7], which assumes nothing about how rewards change. In a *multi-agent* bandit every player is a learner, so each one's target moves as the others learn.

3 Problem Formulation

Stage game $\Gamma_{R,\phi,W,\mathrm{E}}$**,** indexed by the hiring rule R, the utility type ϕ, the network W and the environment level E.

- **Order and types.** The order $1, \dots, n$ is fixed. $s_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$; θ_i is the tercile of s_i, and i observes only θ_i. Candidates thus know roughly how strong they are, but not their score, and interviews occur in a known order.
- **Action.** $S_{i\cdot} \in \{0,1\}^n$ with $S_{ij} = 0$ unless $j \in L(i)$. A candidate can therefore tell only friends who interview after her, so S is filled only above the diagonal and only at friend pairs. In the pilot (groups $\{1,3,5\}, \{2,4,6\}$) exactly six entries are free: $S_{13}, S_{15}, S_{35}, S_{24}, S_{26}, S_{46}$. The last candidate in a friend group has no one to tell and never shares.
- **Leaks.** With $n_j = \sum_{i<j} S_{ij}$ the number of people who tell j: **E0** $k_j = m\mathbb{1}[n_j > 0]$; **E1** $k_j \sim \mathrm{Bin}\big(m, 1 - (1 - m/Q)^{n_j}\big)$; **E2** $k_j = |A_j \cap \bigcup_{i:S_{ij}=1} A_i|$ with $A_j \sim \mathrm{Unif}\binom{[Q]}{m}$. Here k_j is the number of j's questions seen in advance: under E0 a shared question always reappears, under E1 each reappears with the probability that it lies in one of the n_j shared sets, and under E2 the actual questions are drawn from the bank.
- **Score.** $Y_j = s_j + \frac{1}{m}\big(\Delta k_j + \sum_{q=1}^{m} \eta_{jq}\big)$, $\eta_{jq} \sim \mathcal{N}(0, \sigma^2)$, and Y_j is not observed. A score is thus skill, plus marking noise, plus a bonus Δ for each question seen in advance.
- **Hiring.** SEATS: $H = \text{top-}K(Y)$. BAR: $H = \{j : Y_j \geq \tau_K\}$ with $\tau_K = \sqrt{1 + \sigma^2/m}\,\Phi^{-1}(1 - K/n)$. SEATS hires the best K scores, whereas BAR hires everyone above a score chosen to hire the same expected number when nothing leaks.
- **Utilities**, with $|S_{i\cdot}|$ the number of recipients: $u_i^{\text{SELF}} = x_i - c|S_{i\cdot}|$; $u_i^{\text{FRIEND}(\alpha)} = u_i^{\text{SELF}} + \alpha \sum_{j \in F(i)} x_j$; $u_i^{\text{CLAN}(\alpha,\beta)} = u_i^{\text{FRIEND}(\alpha)} - \beta \sum_{j \notin F(i) \cup \{i\}} x_j$; $u_i^{\text{FAIR}(a,b)} = u_i^{\text{SELF}} - \frac{1}{n-1} \sum_j \big[a(x_j - x_i)^+ + b(x_i - x_j)^+\big]$ [13]. Selfish candidates want only their own offer, each recipient told costing c; friendly candidates also value friends' offers at weight α; clannish candidates dislike outsiders' offers at weight β; and fairness-minded candidates dislike inequality.

Level 1 (known environment). Γ is a finite Bayesian game; we seek its BNEs, $\mathcal{E}_{\mathrm{BNE}}(\Gamma)$, and CCEs, $\mathcal{E}_{\mathrm{CCE}}(\Gamma)$. The rules are known, so the question is what a rational candidate does.

Level 2 (unknown environment). Rounds $t = 1, \ldots, T$ are i.i.d. copies of Γ. Candidate i knows her own utility and her arms $\{0,1\}^{|L(i)|}$, but not E, (Q, m, Δ, σ), or the others' utilities and strategies. She observes θ_i^t, chooses $S_{i\cdot}^t$, and sees only u_i^t. The reward of telling a friend is random, depending on who else shares, on whether the question reappears and on everyone's skill, and each candidate must learn it from repeated seasons.

Research questions. RQ1: the equilibrium $S^\star$ and the break-even friendliness $\alpha_{i,\theta}^\star$ (the smallest α at which type θ of candidate i shares), as functions of position, type, W and R. **RQ2:** how utility types ϕ and mixed populations change equilibria. **RQ3:** for each learner, regret, agreement with $\mathcal{E}_{\mathrm{BNE}}$, distance of play from $\mathcal{E}_{\mathrm{CCE}}$, and rounds to reach 90% agreement, across E0–E2. **RQ4:** the company's hiring-value loss $\Lambda_R = \mathbb{E}[V \mid \text{no leaks}] - \mathbb{E}[V]$, with $V = \sum_{j \in H}(s_j - \tau_K)$, as a function of Q.

Structural results. Let ℓ_i be i's own loss in hiring probability from sharing, and G_i the gain in her friends' expected hires.

Lemma 1. *With all random draws and all other rows fixed, setting $S_{ij} = 1$ weakly raises Y_j and changes no other score. Hence $\ell_i \geq 0$ and $G_i \geq 0$ under both rules.*

Telling a friend can only help her, and hence can only hurt the teller and help the teller's friends.

Proposition 1 (No sharing in a clique)**.** *With* SEATS*, complete W,* FRIEND(α)*, $\alpha \leq 1$ and $c > 0$, no one shares in any BNE.*

Proof. $|H| = K$, so $|F(i) \cap H| = K - x_i$ and $G_i = \ell_i$. The net gain from sharing is $\alpha G_i - \ell_i - c = -(1-\alpha)\ell_i - c < 0$. □

If everyone is a friend, every seat gained by a friend is lost by the candidate or by another friend, so sharing is pure cost.

Proposition 2 (Sharing is free under a bar)**.** *Under* BAR *and $\phi \in \{$*SELF, FRIEND, CLAN*$\}$, $\ell_i \equiv 0$ and the gain from any row does not depend on θ_i, so each candidate has a best response that is constant in her type.*

A candidate's offer is settled by her own score, which her friends' scores never affect, so only the friends' gain matters. (For FAIR this can fail, since the penalty depends on whether i herself is hired.)

Proposition 3 (Leak formula)**.** *In E2, each of j's questions is leaked with probability exactly $1 - (1 - m/Q)^{n_j}$. The expected leaked fraction is at most ε if $Q \geq m/(1 - (1-\varepsilon)^{1/n_j})$, and $Q \geq m n_j/\varepsilon$ always suffices.*

Proofs. Lemma 1: Y_j is monotone in k_j, and k_j in n_j. Proposition 2: $x_i = \mathbb{1}[Y_i \geq \tau_K]$ does not depend on S. Proposition 3: independent uniform sets and $(1-\varepsilon)^{1/n} \leq 1 - \varepsilon/n$.

Remark (clannish utilities). Under SEATS, $|H| = K$ gives $\sum_{\text{outsiders}} x_j = K - x_i - \sum_{F(i)} x_j$, so $u_i^{\mathrm{CLAN}(\alpha,\beta)} = (1+\beta)x_i + (\alpha+\beta)\sum_{F(i)} x_j - \beta K - c|S_{i\cdot}|$: up to a constant, a friendly candidate with $\alpha_{\text{eff}} = (\alpha+\beta)/(1+\beta)$, which *rises* with β for $\alpha < 1$. So clannishness should increase sharing under SEATS and do nothing under BAR.

4 Proposed Solutions

Pilot (completed). We used two friend groups, $\{1,3,5\}$ and $\{2,4,6\}$ with $n = 6$, and E1 with $K = 2$, $m = 3$, $Q = 15$, $\Delta = \sigma = 1$, $c = 0.001$ and FRIEND(0.2). The payoff tables (all 3^6 type profiles and all 64 matrices S) are Monte Carlo estimates at 8,000 simulated days per cell. Lemma 1 and propositions 1 to 3 are proved; every pilot number below is a simulation estimate at one parameter setting, not a proof.

- *Equilibrium.* Best-response dynamics from eight starting profiles all reach the same pure BNE, which has zero exploitability *in the estimated table* (its Monte Carlo error is not yet bounded). Under SEATS weak candidates tell all their later friends, while average and strong ones keep quiet; under BAR everyone shares. The break-even $\alpha^\star = (\ell_i + c)/G_i$ is 0.09, 0.32 and 0.37 for weak, average and strong candidates under SEATS, against about 0.01 under BAR.
- *Learning.* A decision cell is a (candidate, type) pair with a real choice. After 60,000 rounds, the most-played row equals the equilibrium row in 72% (SEATS) and 75% (BAR) of cells for ε-greedy, 78% and 97% for UCB1, and 39% and 56% for EXP3. Payoff gaps are about one percentage point against reward noise of order one, so $\sim 10^4$ samples are needed to separate two arms: learning the reward is the core difficulty.

Level 1 methods. (i) Simulate the tables $\mathbb{P}(j \in H \mid \theta, S)$ once per (R, W, E) with common random numbers; every utility type is linear in them, so changing α, β or ϕ needs no new simulation. (ii) Find pure BNEs by best-response enumeration and dynamics from many starts, mixed equilibria by replicator dynamics and support enumeration, and CCEs by a linear program. (iii) Attach a bootstrap ε (Definition 3) to every equilibrium, so "equilibrium" means "ε-equilibrium with a stated ε".

Level 2 methods. Each (candidate, type) pair runs ε-greedy, UCB1, Thompson sampling or EXP3 over its rows. A *model-based* learner instead estimates the leak rate and Δ from the number k_i^t of questions it saw in advance, and best-responds in its estimated game — it learns the environment rather than the values.

Evaluation. *Baselines.* Nobody shares; everybody shares; random sharing; the Level-1 equilibrium (the ceiling for learners); and a leak-free company (the ceiling for the company). *Experiments.* **X1** (core): hiring rule $\times$ $\alpha \in [0, 0.5]$ $\times$ network $W \in \{$two groups, pairs, star, complete$\}$ $\times$ E0–E2, solved at Level 1. **X2** (core): the four utility types and mixed populations, e.g. half selfish and half friendly. **X3** (core): the five learners $\times$ rule $\times$ E0–E2, with $T = 2 \times 10^5$ rounds and 30 seeds each. **X4** (target): the bank size Q swept in E2, giving the company's loss Λ_R. *Metrics.* Sharing rate by position and type; $\alpha^\star$; exploitability ε; hiring-value loss Λ_R; and, for learners, regret, agreement with $\mathcal{E}_{\mathrm{BNE}}$, distance from $\mathcal{E}_{\mathrm{CCE}}$, and rounds to 90% agreement. *Statistics.* 95% bootstrap confidence intervals over seeds, paired comparisons of rules on common random numbers, and Holm correction across hypotheses [14]. *Checks.* Before any experiment the simulator must show that nobody shares at $\alpha = 0$, and reproduce Propositions 1 to 3 and the worked example (0.71, 0.92). In the pilot, $n = 4$ with a complete network gave only the no-sharing equilibrium of Proposition 1. *Budget.* A pilot table took about 90 s and utilities are linear in tables, so X1–X2 need only 24 tables; X3 is 900 runs of 2×10^5 rounds. Estimate: under 10 CPU-hours.

	Hypothesis	Refuted if
H1	Under SEATS, $\alpha^\star_\theta$ is highest for types most likely to be pivotal	$\alpha^\star$ flat, or lowest for pivotal types
H2	BAR shares more than SEATS for every $\alpha \in [0, 0.5]$	SEATS shares more at some α
H3	SEATS: sharing rises with β (CLAN); BAR: β has no effect	no rise under SEATS, or an effect under BAR
H4	No-regret play nears $\mathcal{E}_{\mathrm{CCE}}$; model-based learners are faster	CCE gap does not fall, or not faster
H5	$\Lambda_{\mathrm{BAR}} > \Lambda_{\mathrm{SEATS}}$, and SEATS needs a smaller Q for $\Lambda < 5\%$	CI includes 0, or order reversed

Deliverables and limits. Core: verified simulator, X1–X2 equilibria and CCEs, X3 learning results, verdicts on H1–H4. Target: X4, H5, $n = 8$. Outputs: whom to tell (by position, type, network, rule) and the loss $\Lambda(Q)$. Limits: coarse types, i.i.d. rounds, fixed network, exact solutions only for $n \leq 6$, no real leak data (claims are comparative). *Beyond this proposal (not committed):* a season in which placed candidates leave is a stochastic game [15]. **Timeline:** wks 1–2 simulator, X1; 3 X2; 4 X3; 5 X4; 6 report.

5 Contribution

Jash Bharat Pasad: formulation and proofs; utility types (X2); company analysis (X4); writing and integration. **Bhavay Goyal:** simulator (E0–E2, networks), payoff tables, verification checks, pipeline. **Chaitanya Bhoite:** Level-1 solvers (BNE, mixed equilibria, CCE linear program, bootstrap ε); X1. **Vishnu Sinha:** Level-2 learners (including model-based); regret and CCE metrics; X3; statistics.

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